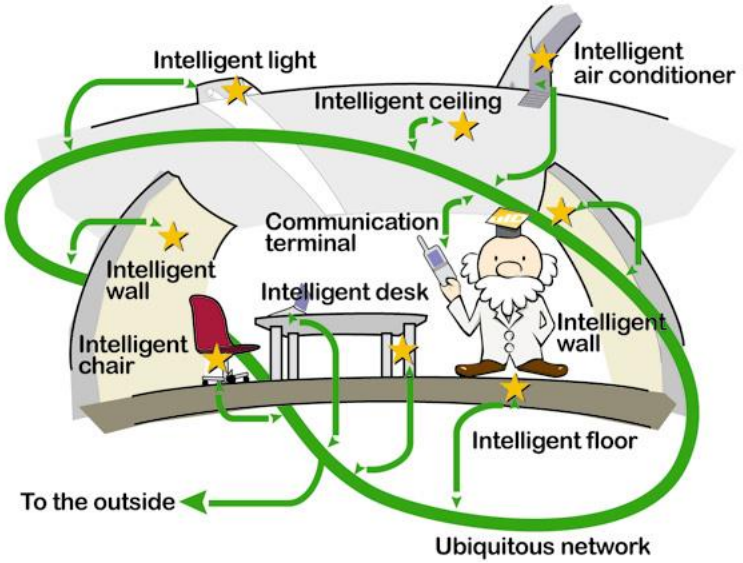
Ubiquitous Computing and Embedded systems

Fernando Guirado
Office 3.17
EPS building



Prototype Development kit

Development kit

- Electronic devices security
- Kit contents

Prototype Development kit

Development kit

- **Electronic devices security statements**
- Kit contents

Safety rules:

- Be careful with the **polarity**!
- All interconnected components have to **share** the **GND line**
- **First** - wires connection, **Next** - power supply
- Electronic devices vs Static electricity 😱
- Wires are so fragile 😓
- Some components requires a stable power supply
- Some devices require **3.3V** and others **5V – Pay attention!** 🤯
  (Arduino, Raspberry, NodeMCU, ESP-01, …)

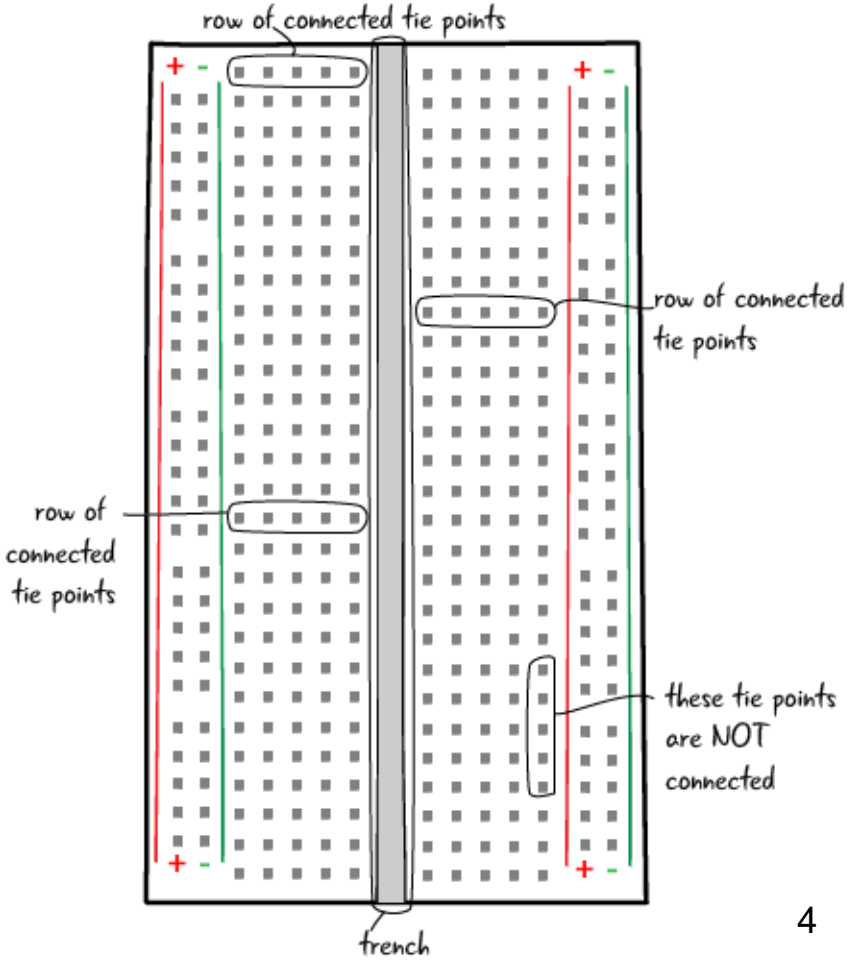
Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**

Breadboards



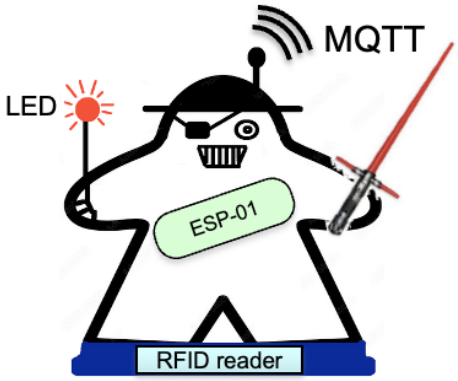


Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**

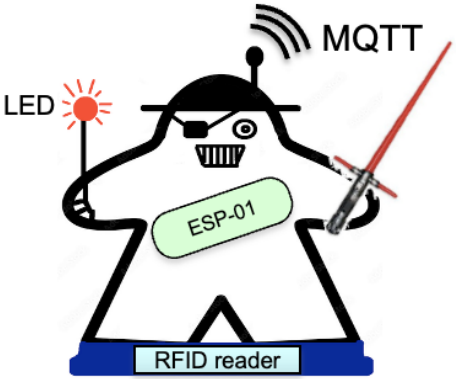
Meeple Components



Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**



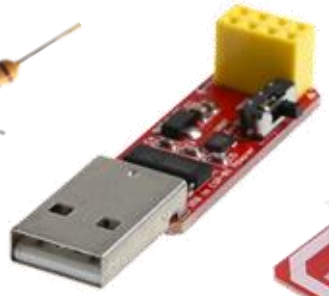
Meeple

Each kit contains:

- 4 x ESP-01
- 4 x ESP-01 Programmer module
- 4 x ESP-01 Bredboard adapter
- 4 x Mini - Bredboard
- 4 x PN532 RFID Reader
- 4 x RFID TAG
- LEDs + Resistors
- Set of Wires










RFID Reader
PN532



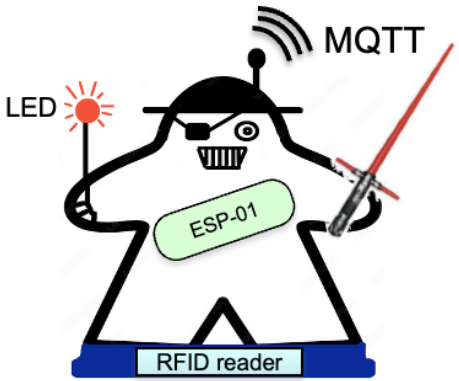




Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**

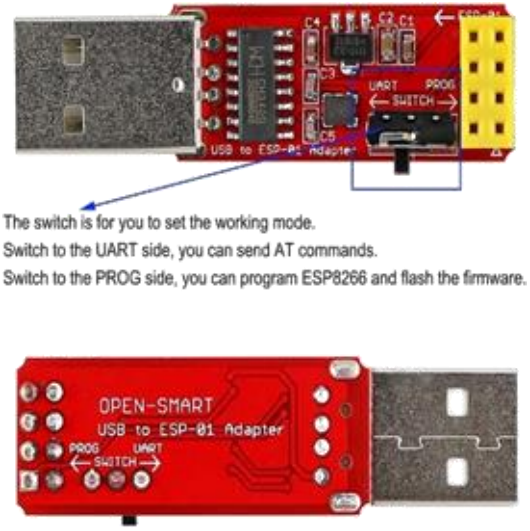


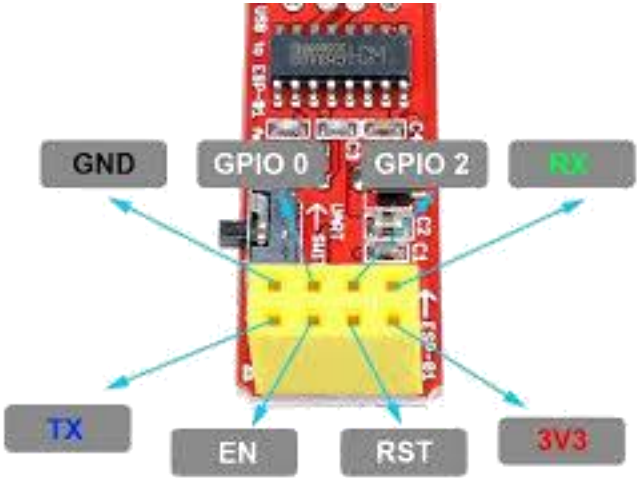
Meeple

Programming the ESP-01

- It is 5v tolerant
- Unplug previous to select the switch mode position



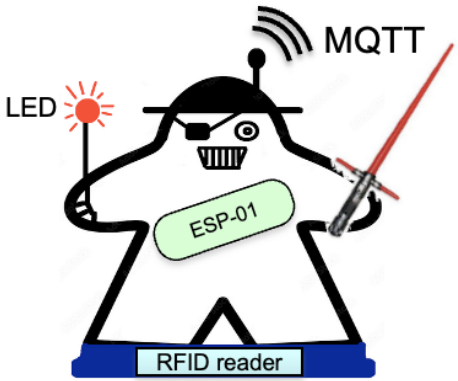




Prototype Development kit

Development kit

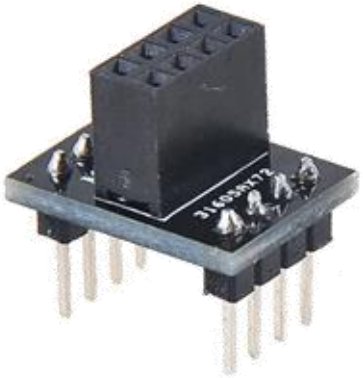
- Electronic devices security
- **Kit contents**



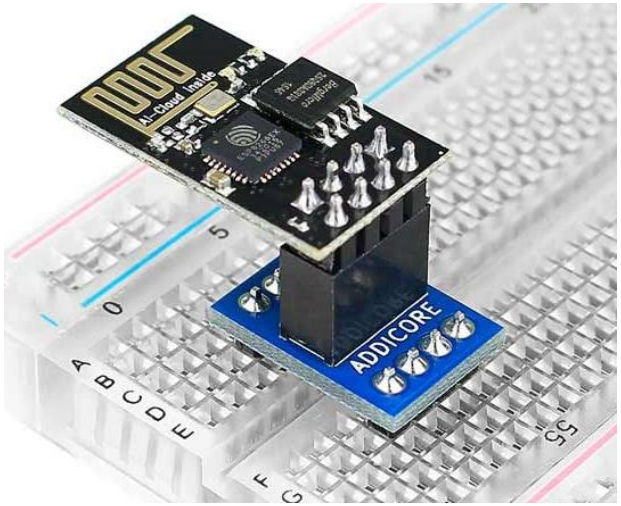
Meeple

ESP-01 – Breadboard adapter

- It is 5v tolerant
- Unplug previous to select the switch mode position



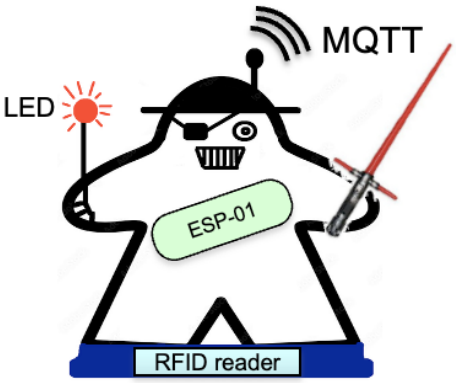




Prototype Development kit

Development kit

- Electronic devices security
- Kit contents

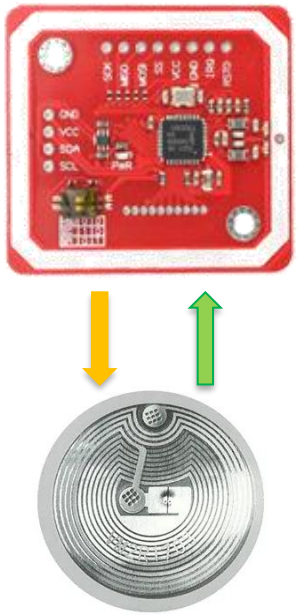


Meeple

RFID Reader PN532

An **RFID reader** is an electronic device that emits radio-frequency waves to detect, read, and transmit information stored on an **RFID tag** or card without the need for direct line of sight.

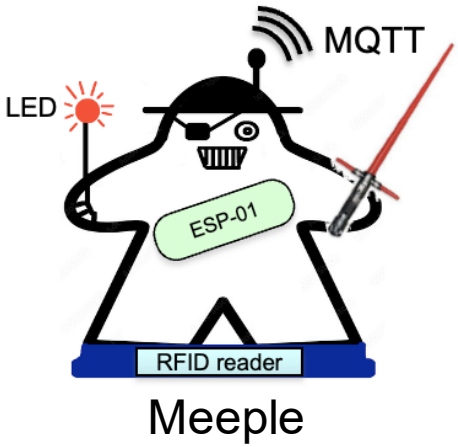
1. **Signal emission:** The reader transmits electromagnetic waves via an antenna.
2. **Activation:** The tag (containing a microchip and an antenna) captures that energy and powers up.
3. **Response:** The tag transmits its encoded data back to the reader.
4. **Processing:** The reader receives the information and sends it to the microcontroller to process data.



Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**

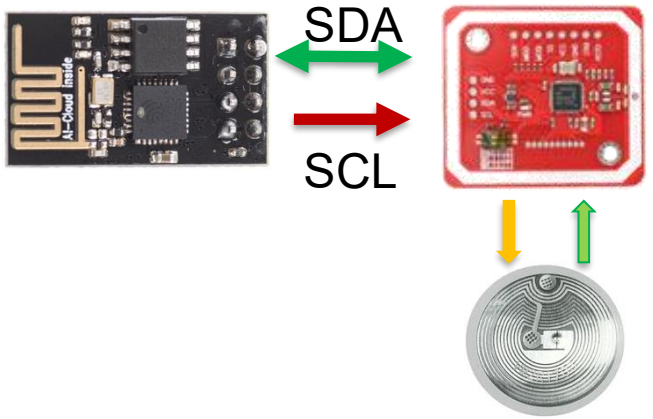


Meeple

RFID Reader PN532

It features three communication interfaces:

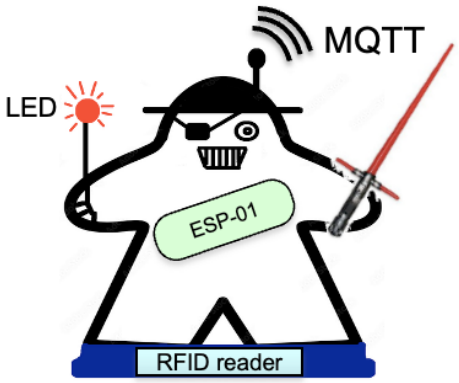
- **I2C:** uses two data/clock lines (SDA and SCL)
- **SPI:** offers maximum data transfer speed; recommended for projects with high processing loads.
- **HSU (High Speed UART):** high-speed direct serial connection, useful for communicating with a PC or boards with dedicated serial ports.



Prototype Development kit

Development kit

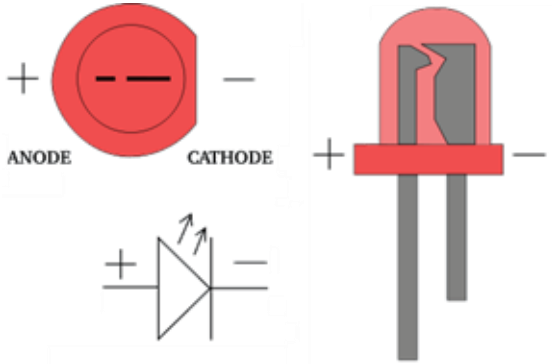
- Electronic devices security
- Kit contents

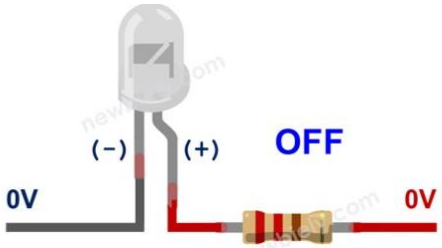


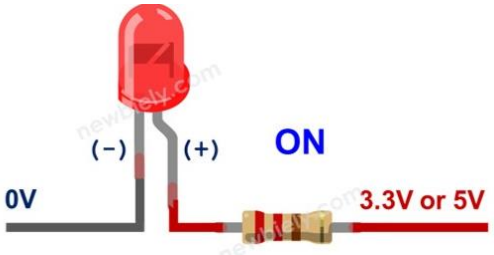
Meeple

LED

A light-emitting diode (LED) is semiconductor device that emits light when current flows through it.

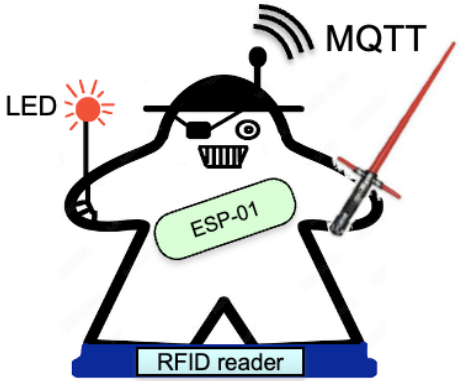






Prototype Development kit

Proposed Schema



Meeple



Powerbank

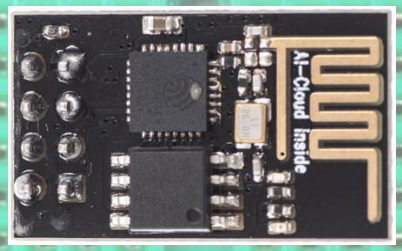
ESP01 Adapter

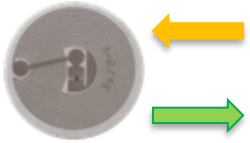


VCC

GND

ESP01





RFID Tag



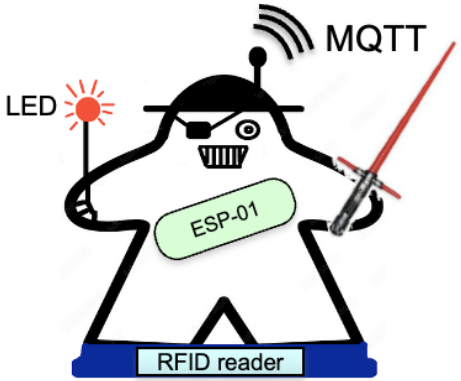
RFID Reader PN532

I2C Bus

LED

Prototype Development kit

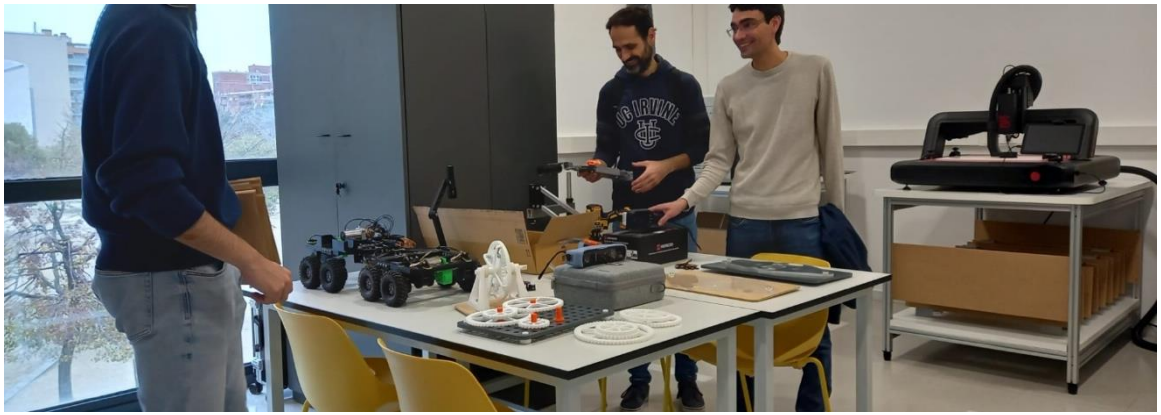
Proposed Schema



Meeple

It is possible to access the EPS MakerLAB to create an "exoskeleton" for the meeple.

- 3D printers
- Milling Machine
- Laser Cutter and Engraver
- ...



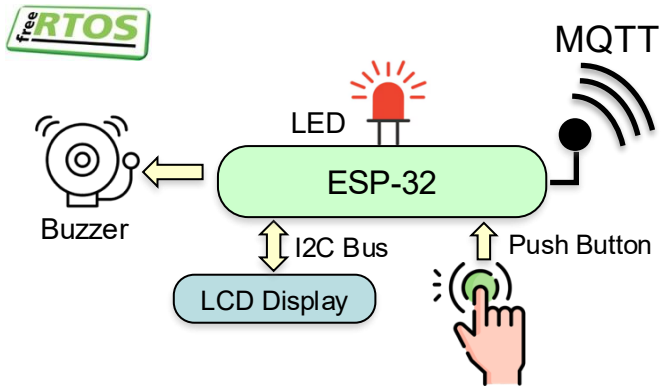
https://makerslab.udl.cat/ca/

Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**

Control Base Components

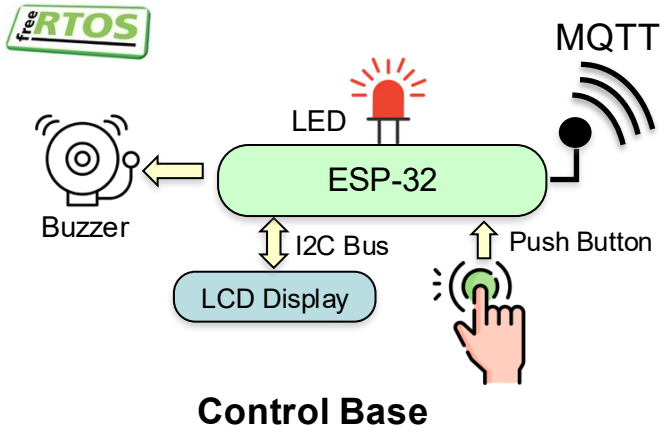


Control Base

Prototype Development kit

Development kit

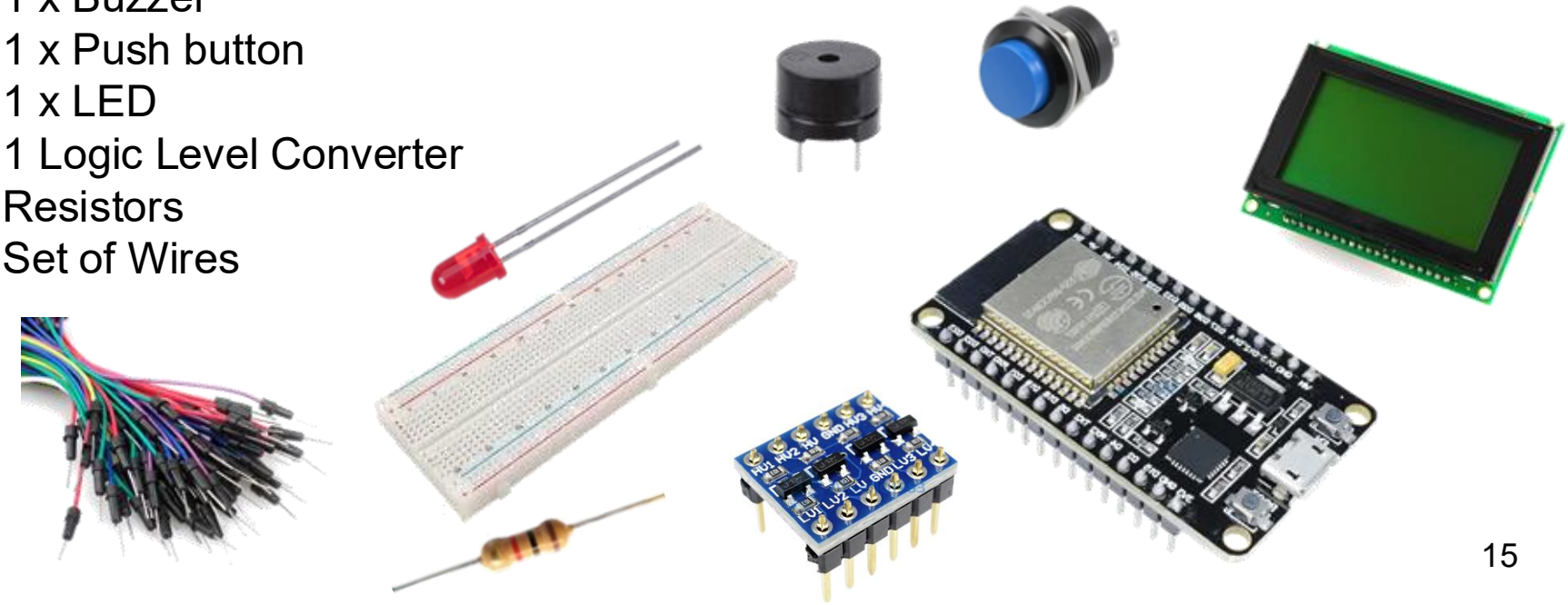
- Electronic devices security
- **Kit contents**



Control Base

Each kit contains:

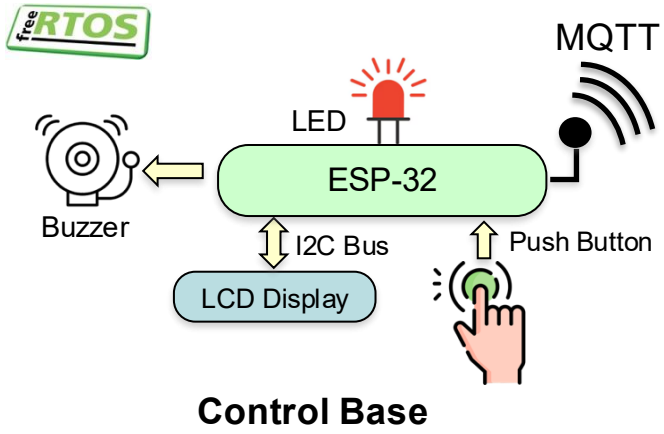
- 4 x ESP-32
- 1 x Graphic LCD Display (128x64 pixels)
- 1 x Buzzer
- 1 x Push button
- 1 x LED
- 1 Logic Level Converter
- Resistors
- Set of Wires



Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**



Control Base

Important!

The ESP32 has a **micro-USB** connector.

You must provide the USB cable to program it.

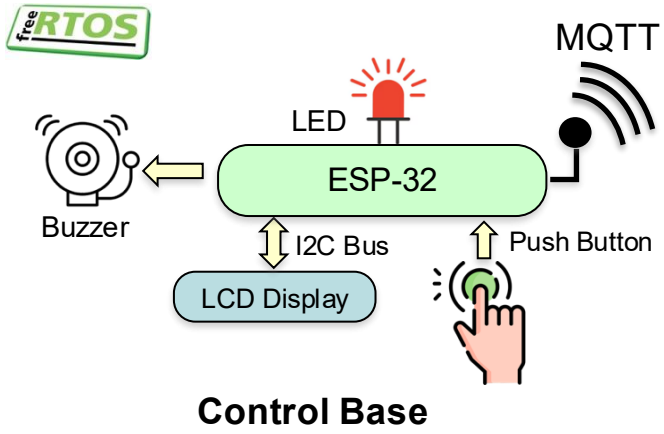




Prototype Development kit

Development kit

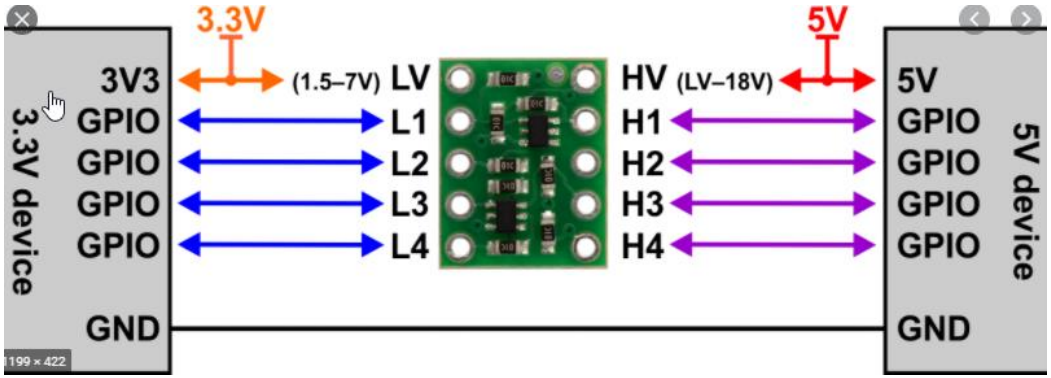
- Electronic devices security
- **Kit contents**



Control Base

Logic Level Converter

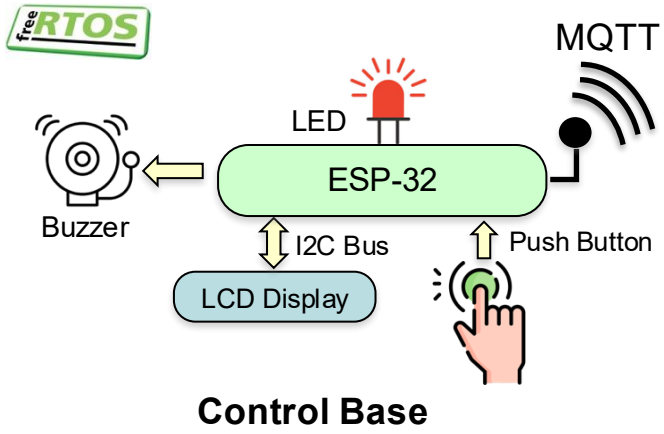
A **logic level converter** is a essential component for safely connecting devices that operate at different voltages (typically 5V and 3.3V), protecting your microcontroller pins from potential overvoltage damage.



Prototype Development kit

Development kit

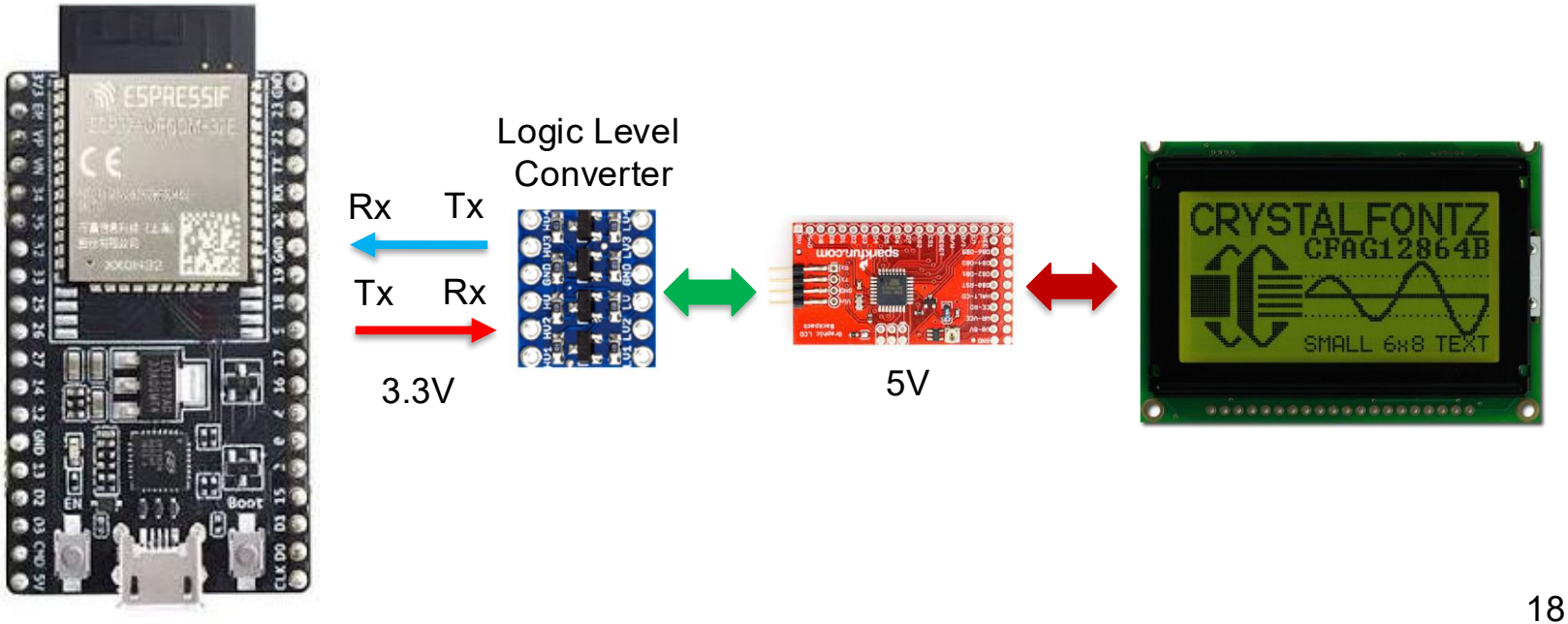
- Electronic devices security
- **Kit contents**



Control Base

Graphic Serial LCD Display

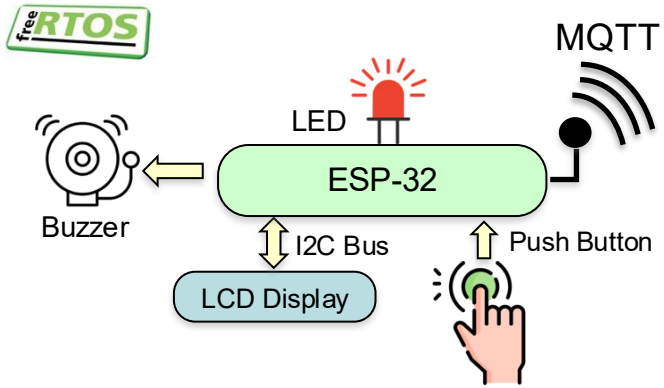
- Serial Transmission – TX/RX



Prototype Development kit

Development kit

- Electronic devices security
- **Kit contents**



Control Base

Piezo Buzzer (passive)

A passive piezo buzzer is an audio signaling device that generates sound when it is activated by a square wave.

```
void tone(pin, frequency)
void tone(pin, frequency, duration)
```

Generates a **square wave** of the specified frequency on a pin. A duration can be specified.

It uses timers depending on how many simultaneous notes you wish to use.



3-5V Buzzer

- Connect directly to ESP32 => normal sound

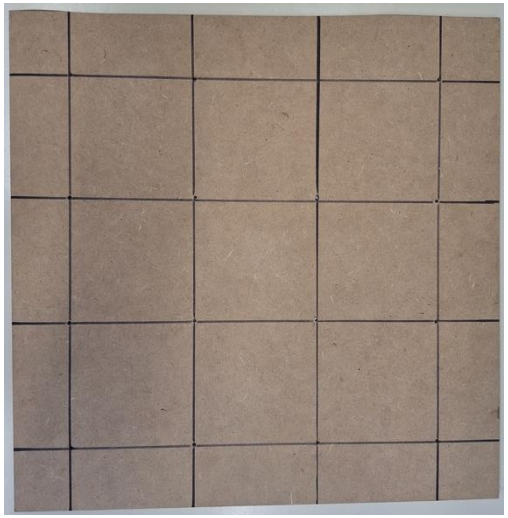
Development kit

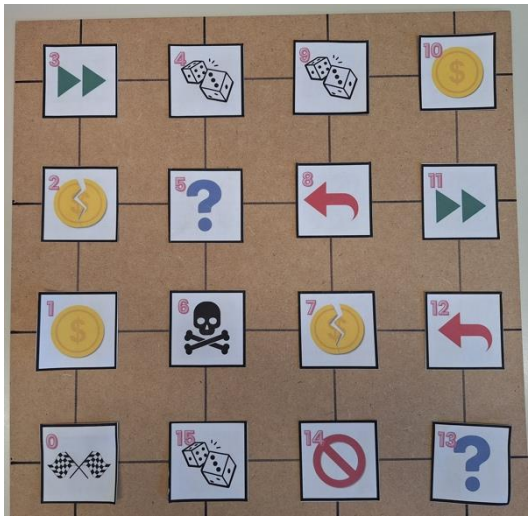
Development kit

- Electronic devices security
- **Kit contents**

Game Board

consisting of a 4x4 grid of cells, each identified by an RFID tag







Development kit

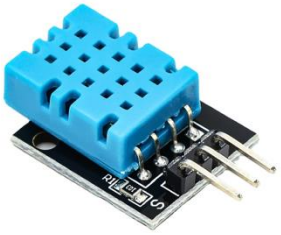
Development kit

- Electronic devices security
- Kit contents
  - Additional components



These components will be used to carry out individual activities based on the ESP32.

- **DHT11 – Temperature & Humidity Sensor**
- **HCSR04 / HCSR03 – UltraSonic Distance Sensor**
- **ADXL345 – Accelerometer Sensor**



DHT11



HCSR04



HCSR03



ADXL345

Prototype Development kit

Development environment

- Is up to you which environment to use:
  - Standalone Arduino IDE
  - Visual Studio + Arduino IDE, PlatormIO, Espressif IDE



- It is necessary to create a documented code repository to track the project.
- There are some simulators that can help you verify your code before moving on to the physical prototype.

wokwi.com

www.cirkitdesigner.com

velxio.dev

picsimlab.com

Prototype Development kit

Take care about

- It would be preferable to use your own laptop.
  It will be necessary to install the IDE, libraries, cross-compilers, etc.
- An ad hoc LAN must be created to allow MQTT access.
- It is necessary some USB cables to program the microcontrollers.
- Some devices can share power supply, BUT! be extremely careful when doing so. Any short circuit could damage not only the components but also your laptop.
- Some components are very sensitive to the switch ON / OFF process! (ESP01 & ESP32 require a minimum amount of time to start-up)

Prototype Development kit

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Enjoy yourself and don't stress!